they could readily distinguish features related to clouds in the atmosphere from features related to the surface of Earth itself.

Time for a reality check. The nearest exoplanet—the nearest planet in orbit around a star that is not the Sun—can be found in our neighbor star system Alpha Centauri, about four light-years from us and visible mostly from our southern hemisphere. Most of the cataloged exoplanets lie from dozens up to hundreds of light-years away. Earth's brightness is less than one-billionth that of the Sun, and our planet's proximity to the Sun would make it extremely hard for anybody to see Earth directly with a visible light telescope. It's like trying to detect the light of a firefly in the vicinity of a Hollywood searchlight. So if aliens have found us, they are likely looking in wavelengths other than visible light, like infrared, where our brightness relative to the Sun is a bit better than in visible light —or else their engineers are adapting some other strategy altogether.

Maybe they're doing what some of our own planet-hunters typically do: monitoring stars to see if they jiggle at regular intervals. A star's periodic jiggle betrays the existence of an orbiting planet that may be too dim to see directly. Contrary to what most people suppose, a planet does not orbit its host star. Instead, both the planet and its host star revolve around their common center of mass. The more massive the planet, the larger the star's response must be, and the more measurable the jiggle gets when you analyze the star's light. Unfortunately for planet-hunting aliens, Earth is puny, so the Sun barely budges, which would further challenge alien engineers.

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NASA's Kepler telescope, designed and tuned to discover Earth-like planets around Sun-like stars, invoked yet another method of detection, mightily adding to the exoplanet catalog. Kepler searched for stars whose total brightness drops slightly, and at regular intervals. In these cases, Kepler's line of sight is just right to see a star get dimmer, by a tiny fraction, due to one of its own planets crossing directly in front of the host star. With this method, you can't see the planet itself. You can't even see any features on the star's surface. Kepler simply tracked changes in a star's total light, but added thousands of exoplanets to the catalog, including hundreds of multiplanet star systems. From these data, you also learn the size of the exoplanet, its orbital period, and its orbital distance from the host star. You can also make an educated inference on the planet's mass.